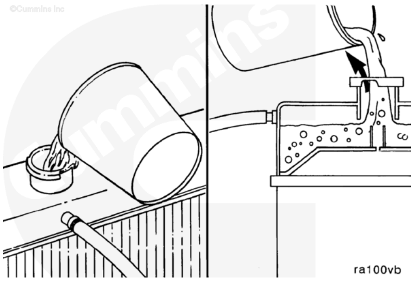
Trainings

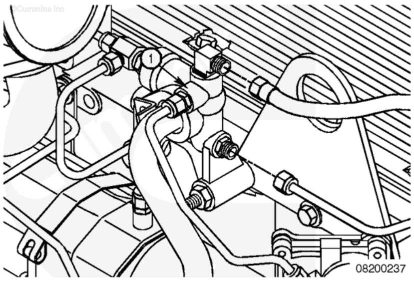
Initial Check

The system **must** be designed to allow air to escape while filling.

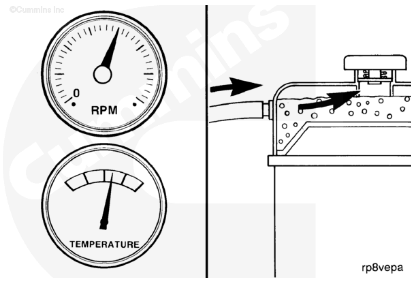


The cooling system vent line is routed from the water manifold (1) to the radiator fill or auxiliary tank above the coolant level.

Use a Number 4 flexible hose.



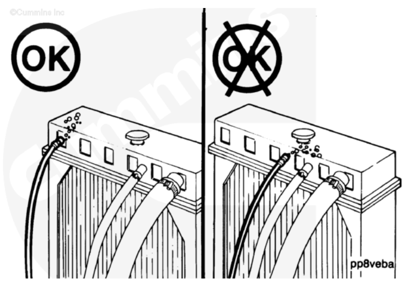
During engine operation, coolant will continue to flow through the engine vent line to remove air from the coolant.



Do **not** route the fill or vent line in a manner that will allow air to be trapped in the system.

Connect the vent line to the radiator fill/auxiliary tank, as far away from the fill line as possible, to prevent turbulence in the fill line.

Route the vent line with a continuous rise so that coolant and air can continue to flow after engine shutdown.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

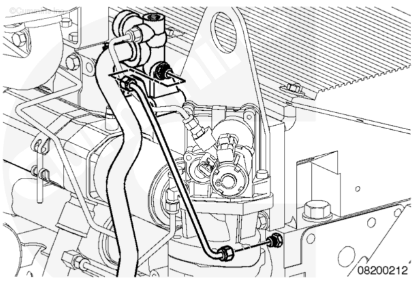
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the coolant system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the thermostat housing. Refer to Procedure 008-013 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-013-tr.html)
- Loosen (do **not** remove) the water manifold mounting capscrews. See equipment manufacturer service information.

Remove

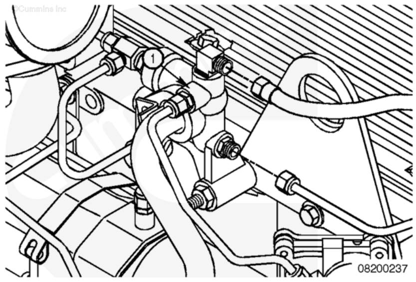
Note : Do not remove the EGR valve in order to gain access to the cylinder head vent line.

Remove the cylinder head vent line from the cylinder head and the water manifold.

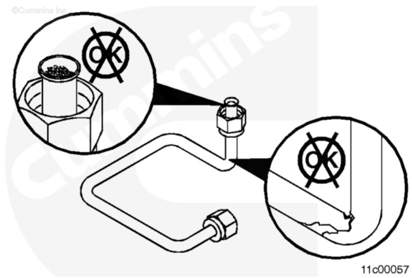


Disconnect the coolant vent line from the water manifold and the radiator.

Remove the vent line.



Clean and Inspect for Reuse



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the vent lines with solvent.

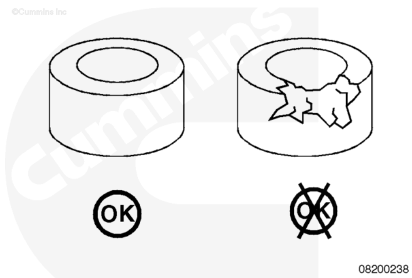
Dry with compressed air.

Inspect the cylinder head vent line and coolant vent line for cracks or other damage.

Replace the cylinder head vent line or coolant vent line if cracks or other damage is found.

Inspect the grommets for cracks or other damage.

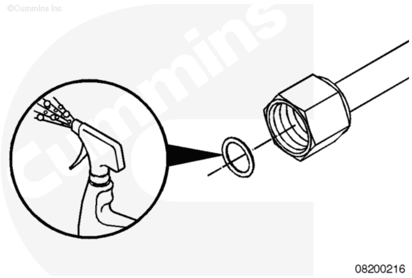
Replace the grommets if damage is found.



Install

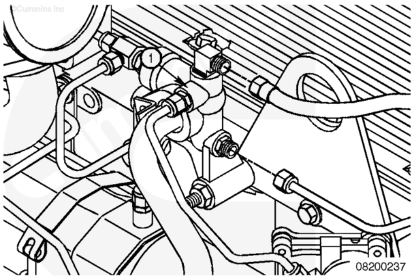
⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing grommets. The grommets will swell and cause the connection to leak. Lubricate the grommets with clean engine coolant, soapy water, or vegetable oil to avoid damaging the groummets while assembling.



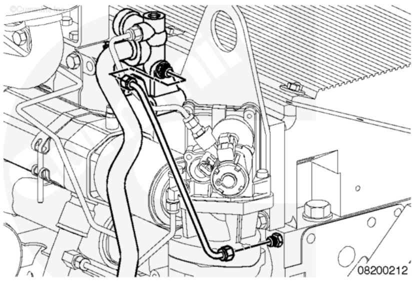
Lubricate the grommets with clean engine coolant, soapy water, or vegetable oil to avoid damaging the grommets during assembly.

Install the coolant vent line from the water manifold to the radiator.



Install the cylinder head vent line from the cylinder head to the water manifold.

Torque Value: 14 n•m [124 in-lb]



Finishing Steps

- Tighten the water manifold mounting capscrews. Refer to Procedure 008-061 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Install the thermostat housing. Refer to Procedure 008-013 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-013-tr.html)

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine until it reaches a temperature of 82°C [180°F] and check for coolant leaks.

Last Modified: 01-Apr-2015